

- 2 The high blood pressure at the arterial end of a capillary network results in some components of blood leaving the capillaries and forming tissue fluid. At the venous end, the presence of plasma proteins allows movement of water by osmosis back into the capillaries.

Fig. 2.1 is a diagram showing a capillary network. The lymph vessels and the blood vessels at the arterial and venous ends of the network are also shown.

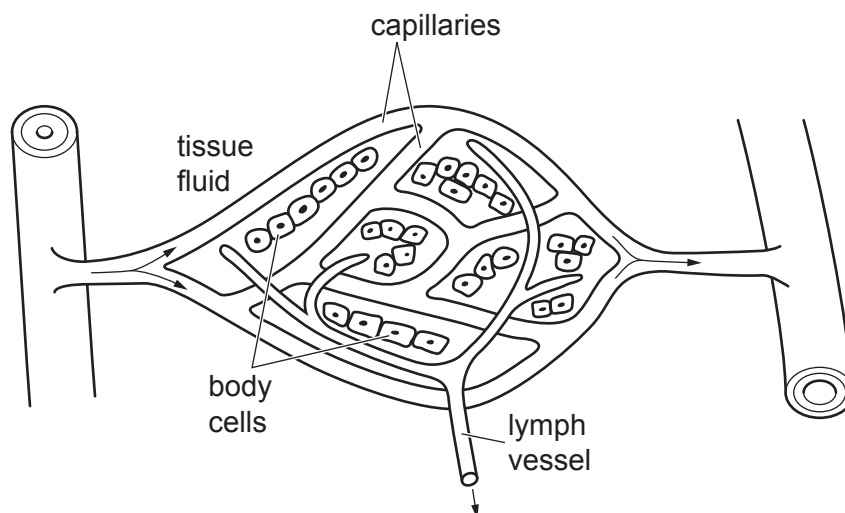


Fig. 2.1

- (a) Red blood cells and plasma proteins, such as albumin, remain in the capillaries and are not found in tissue fluid.

Explain why red blood cells and albumin do not leave the capillary.

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..... [2]

- (b) Name the chemical reagent or reagents used to test for proteins in a sample of blood plasma **and** state the colour change that will be seen if protein is present.

reagent or reagents

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colour change

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[2]

- (c) When a person moves from sea level and stays at high altitude for a few months, there is an increase in the number of red blood cells per mm^3 of blood.

Explain why this increase occurs.

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- (d) A low blood albumin concentration can lead to a condition known as oedema.

Oedema is a swelling of tissues caused by the accumulation of tissue fluid surrounding the body cells in the capillary network.

Suggest **and** explain how a low blood albumin concentration can lead to oedema.

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- (e) Albumin transports some cell signalling molecules from the cells where they are synthesised to their target cells. The cell signalling molecules bind to specific cell surface membrane proteins on the target cells.

Name the type of membrane protein to which the cell signalling molecules bind.

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[Total: 11]